



EXAM PAPERS PRACTICE

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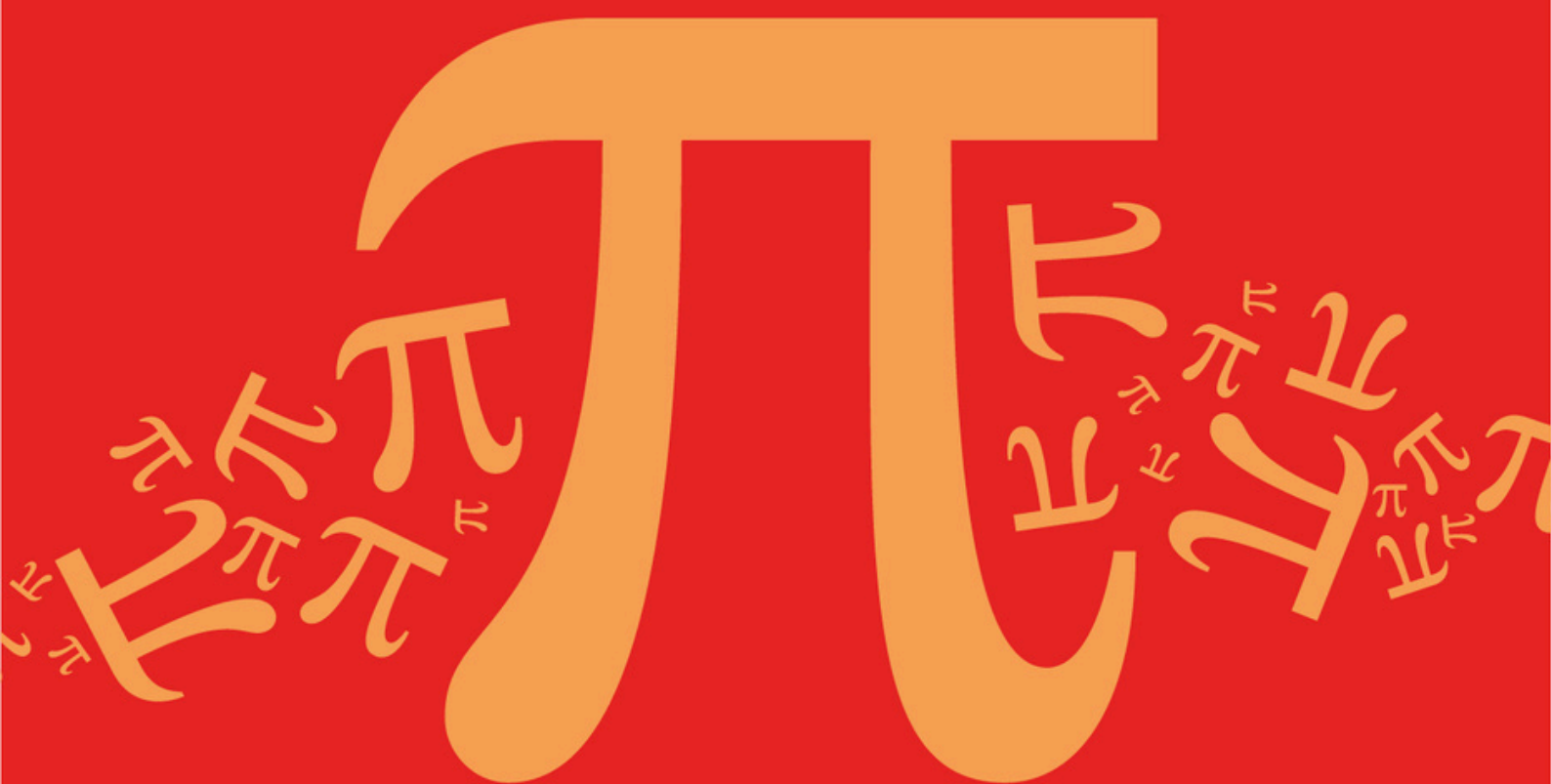
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Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C7: Inverse Matrices to Solve Simultaneous Equations

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C6: Singular Matrices (Properties of Singular Matrices)

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

(a) Given that $\mathbf{A}^2 = \begin{bmatrix} p & -2 & -4 \\ 5 & 6 & 4 \\ 10 & q & 9 \end{bmatrix}$, find the value of p and the value of q .

(2)

(b) Given that $\mathbf{A}^3 - 6\mathbf{A}^2 + 11\mathbf{A} - 6\mathbf{I} = \mathbf{0}$, prove that

$$\mathbf{A}^{-1} = \frac{1}{6}(\mathbf{A}^2 - 6\mathbf{A} + 11\mathbf{I})$$

(2)

(c) Given that $\mathbf{A}^{-1} = \frac{1}{6} \begin{bmatrix} r & -2 & 2 \\ -1 & 5 & -2 \\ -2 & s & 2 \end{bmatrix}$, find the value of r and the value of s .

(2)

(d) Hence, or otherwise, find the solution of the system of equations

$$x - z = k$$

$$x + 2y + z = 5$$

$$2x + 2y + 3z = 7$$

giving your answers in terms of k .

(3)

(Total 9 marks)

Q2.

The system of equations

$$2x - y - z = 3$$

$$x + 2y - 3z = 4$$

$$2x + y + az = b$$

does not have a unique solution.

(a) Show that $a = -3$.

(3)

(b) Given further that the equations are inconsistent, find the possible values of b .

(2)

(Total 5 marks)

Q3.

- (a) Determine the two values of the integer n for which the system of equations

$$2x + ny + z = 5$$

$$3x - y + nz = 1$$

$$-x + 7y + z = n$$

does not have a unique solution.

(4)

- (b) For the positive value of n found in part (a), determine whether the system is consistent or inconsistent, and interpret this result geometrically.

(6)

(Total 10 marks)